

$$\phi_2 = a_x^2 + a_y^2 - \frac{(a_x b_x + a_y b_y)^2}{b_x^2 + b_y^2}.$$

To calculate the curve-averaged value of the basis function of the p^2 polynomial in Eq. 19, the following integral needs to be calculated

$$\int_0^1 (\xi + \phi_1)^p \sqrt{\phi_0(\xi + \phi_1)^2 + \phi_2} d\xi = \left(\frac{\phi_2}{\phi_0}\right)^{\frac{p+1}{2}} \sqrt{\phi_2} \int_{\theta_0}^{\theta_1} \tan^p \theta \sec^3 \theta d\theta, \quad (37)$$

where $\theta_0 = \arctan(\sqrt{\frac{\phi_0}{\phi_2}} \phi_1)$, $\theta_1 = \arctan(\sqrt{\frac{\phi_0}{\phi_2}} (1 + \phi_1))$. If p is an odd integer $p = 2k + 1$,

$$\int \tan^{2k+1} \theta \sec^3 \theta d\theta = \int (\sec^2 \theta - 1)^k \sec^2 \theta d\sec \theta. \quad (38)$$

If p is an even integer $p = 2k$,

$$\int \tan^{2k} \theta \sec^3 \theta d\theta = \int (\sec^2 \theta - 1)^k \sec^2 \theta d\theta, \quad (39)$$

and

$$I_n = \int \sec^n \theta d\theta = \frac{1}{n-1} (\tan \theta \sec^{n-2} \theta + (n-2) I_{n-2}).$$

B. Quadratic integral on curved triangle

The isoparametric transformations of the quadratic triangular cell Eq. 30 is

$$\begin{aligned} \frac{\partial x}{\partial \xi} &= a_0 \xi + a_1 \eta + a_2, \\ \frac{\partial x}{\partial \eta} &= a_3 \xi + a_4 \eta + a_5, \\ \frac{\partial y}{\partial \xi} &= b_0 \xi + b_1 \eta + b_2, \\ \frac{\partial y}{\partial \eta} &= b_3 \xi + b_4 \eta + b_5, \end{aligned} \quad (40)$$

where $a_0 = 4(x_0 + x_1 - 2x_3)$, $a_1 = 4(x_0 - x_3 + x_4 - x_5)$, $a_2 = -3x_0 - x_1 + 4x_3$, $a_3 = 4(x_0 - x_3 + x_4 - x_5)$, $a_4 = 4(x_0 + x_2 - 2x_5)$, $a_5 = -3x_0 - x_2 + 4x_5$, $b_0 = 4(y_0 + y_1 - 2y_3)$, $b_1 = 4(y_0 - y_3 + y_4 - y_5)$, $b_2 = -3y_0 - y_1 + 4y_3$, $b_3 = 4(y_0 - y_3 + y_4 - y_5)$, $b_4 = 4(y_0 + y_2 - 2y_5)$, $b_5 = -3y_0 - y_2 + 4y_5$.